

AI ML ASSIGNMENT - BOTBRAIN

WEEK 3 & WEEK 4 REPORT

WEEK 3 REPORT

1. Chatbot Interface & NLP

What I did: Developed a chatbot interface inside the Streamlit frontend with a floating toggle button and popup. Integrated Google Gemini Flash API with the FastAPI backend (`/chat` endpoint). Implemented a custom text-processing function (`extract\_navigation\_request`) using regular expressions to identify 'from' and 'to' locations in user messages.

How I met the requirement: The chatbot can handle free-text queries like “Navigate from Hostel to Admin Block” and return meaningful responses.

2. Navigation Module

What I did: Built the backend `/navigate` endpoint that supports BFS, DFS, UCS, and A*. Added distance calculation and estimated walking time (`total\_distance`, `estimated\_time`). Implemented a function to retrieve building info dynamically for each node in the path.

How I met the requirement: Users can select any two locations from dropdowns and choose a search algorithm. The system displays path details (distance, time, and building info). Results are also plotted on an interactive Folium map in Streamlit.

3. Data Handling

What I did: Represented the campus as a graph using building nodes and weighted edges. Used Python dictionaries for storing coordinates and metadata (like building descriptions).

How I met the requirement: The system can query building data, retrieve coordinates, and display location info. Successfully provides visual markers and path overlays on the map

```
LOCATIONS = [
    {"name": "Main Gate", "lat": 13.221306, "lng": 77.755056},
    {"name": "Admin Block", "lat": 13.222194, "lng": 77.755250},
    {"name": "Academic Block 1", "lat": 13.223333, "lng": 77.754917},
    {"name": "Academic Block 2", "lat": 13.223389, "lng": 77.756056},
    {"name": "Academic Block 3", "lat": 13.222389, "lng": 77.756278},
    {"name": "Hostel", "lat": 13.224528, "lng": 77.758056},
    {"name": "Food Court", "lat": 13.224861, "lng": 77.757222},
    {"name": "Sports Complex", "lat": 13.228417, "lng": 77.758278},
    {"name": "Central Junction", "lat": 13.222778, "lng": 77.755611},
    {"name": "Laundry", "lat": 13.224528, "lng": 77.757056},
    {"name": "Mini Mart", "lat": 13.224556, "lng": 77.758028},
    {"name": "Generator Room", "lat": 13.223210, "lng": 77.757350},
    {"name": "Small Junction", "lat": 13.224850, "lng": 77.758410},
]
```

Deliverables

- Integrated Backend + Frontend (FastAPI + Streamlit)
 - Navigation System with BFS, DFS, UCS, and A*
 - Working Chatbot powered by Gemini
 - Map Visualization with Folium markers and paths
 - Space for attaching source code snapshots here:
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WEEK 4 REPORT

Testing & Finalization

1. Testing & Accuracy

What I did: Ran multiple navigation queries across all 12+ campus locations. Validated distances and routes for correctness. Compared algorithms on efficiency and node expansions.

How I met the requirement: BFS and DFS provided valid paths but were less efficient. UCS and A* consistently produced the shortest distance paths. A* expanded fewer nodes, proving to be the best-performing algorithm.

2. UI Refinement

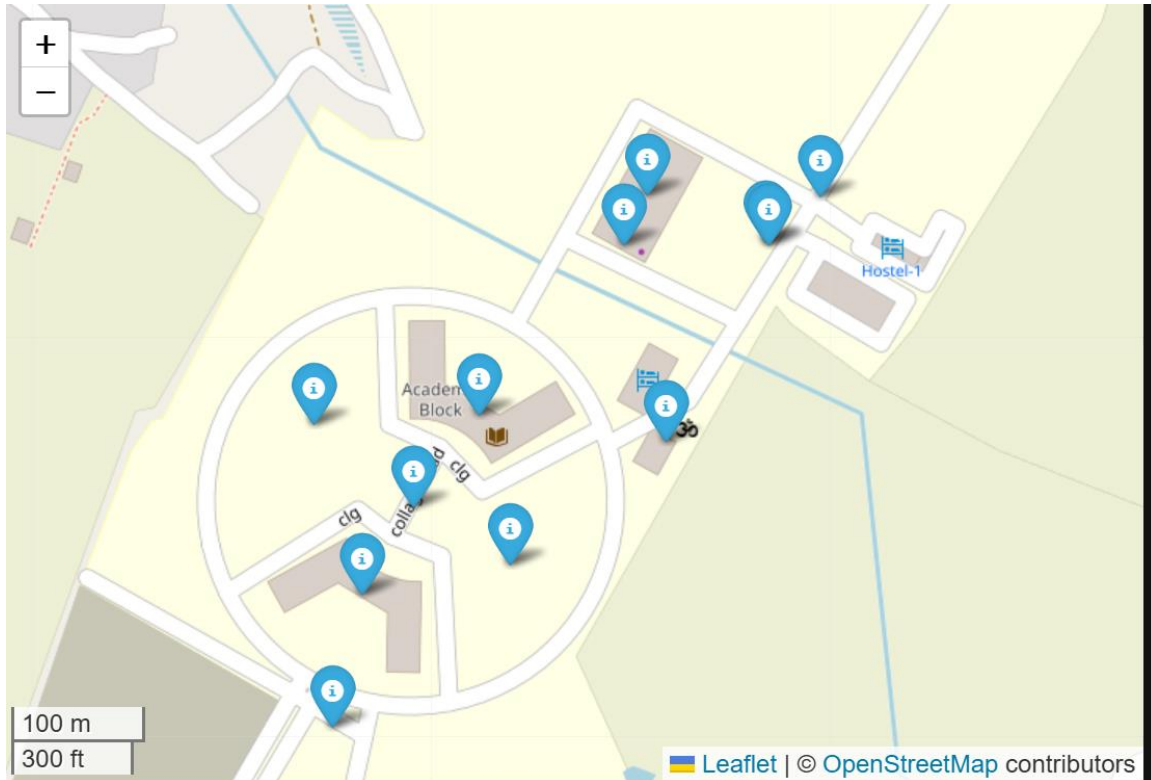
What I did: Styled the Streamlit frontend with a dark theme. Added error handling. Improved output formatting with highlighted navigation details.

How I met the requirement: Clear, user-friendly UI with dropdowns for path selection. Floating chatbot toggle with conversation history. Dynamic error messages and status indicators improve usability.

3. Final Report & Documentation

What I did: Prepared full technical documentation including PEAS analysis, system architecture, and algorithm explanations. Added requirements.txt files for both backend and frontend. Created a clear README for GitHub submission.

How I met the requirement: Deliverables (code, docs, demo) match assignment requirements. Presentation prepared with system overview, algorithm demonstration.



Deliverables

- Fully Functional Prototype (Campus navigation + chatbot)
- Algorithm Comparison (BFS, DFS, UCS, A*)
- Documentation (Final Report, README, requirements)
- Working Demo ready for submission